WIL Data Analyst: Data Engineering and Preprocessing Report

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1 Introduction

This report documents the preprocessing pipeline for a customer churn prediction project using artificial neural networks. The dataset used is `Dataset_with_IsNewCustomer.csv`, which contains customer attributes including service configuration, tenure, and billing information. The objective of preprocessing is to transform raw data into a structured format suitable for machine learning modelling and further analytical tasks.

Preprocessing is a critical stage because raw data often contains inconsistencies, scale imbalance, and categorical complexity. A well-designed preprocessing pipeline improves model stability and ensures that the learning algorithm focuses on meaningful patterns rather than noise.

2 Feature Engineering

New Customer Identification

The `tenure` feature represents the number of months a customer has been with the service. Customers with very low tenure are typically new and may exhibit unstable behaviour patterns. Therefore, a new binary feature `IsNewCustomer` was created:

$$\texttt{IsNewCustomer} = \begin{cases} 1, & \texttt{if tenure} \leq 2 \\ 0, & \texttt{otherwise} \end{cases}$$

This transformation enables the model to capture early-stage customer behaviour more effectively. The feature also introduces a behavioural segmentation dimension that cannot be directly observed from raw tenure values.

Label Positioning

The target variable `Churn` was moved to the last column. This ensures compatibility with machine learning frameworks and improves pipeline consistency. This arrangement also simplifies feature-label separation during model training.

3 Preprocessing Strategy

Numerical Features

Continuous variables were normalised using Min-Max scaling:

$$x_{\text{norm}} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

This transformation ensures that all numerical features share a comparable scale. It is particularly important for distance-based methods and neural networks. Without scaling, features with larger ranges could dominate the learning process and distort model behaviour.

Categorical Features

Categorical variables were processed using two strategies:

- Contract type was one-hot encoded to preserve category independence.
- Other categorical features were converted into numerical codes to reduce dimensionality.

This hybrid approach balances interpretability and efficiency. One-hot encoding preserves structural meaning for contract types. Simple encoding reduces feature explosion for binary variables.

4 Scaling Techniques Documentation

Overview

Scaling is a fundamental preprocessing step in machine learning pipelines. It ensures that numerical features contribute equally during model training. Without scaling, features with larger ranges may dominate distance calculations and gradient updates.

In this project, Min-Max scaling was applied to all continuous numerical variables. This approach transforms features into a fixed range between 0 and 1.

Min-Max Scaling Formula

The Min-Max scaling transformation is defined as:

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

where x represents the original value, $x_{\min}$ is the minimum value of the feature, and $x_{\max}$ is the maximum value of the feature.

This transformation preserves the relative distribution of the data while normalising its range.

Purpose of Min-Max Scaling

Min-Max scaling was selected for the following reasons:

- It ensures that all features lie within the same numerical range.
- It improves convergence behaviour in neural network training.
- It prevents dominance of high-magnitude variables in distance-based methods such as K-Means.
- It preserves the original distribution shape of the data.

This makes it particularly suitable for both ANN modelling and clustering analysis.

Application in This Project

The following features were scaled using Min-Max scaling:

- `tenure`
- `MonthlyCharges`
- `SeniorCitizen`

Scaling was applied after feature engineering and before model training. This ensures consistency across both supervised and unsupervised learning tasks.

Code Implementation

The preprocessing and scaling pipeline was implemented using Python with the `pandas` and `scikit-learn` libraries. The following code summarises the main steps used in this project.

```
import pandas as pd
from sklearn.preprocessing import MinMaxScaler

# Load dataset
df = pd.read_csv('Dataset_with_IsNewCustomer.csv')

# Ensure label is placed at the end
label_column = 'Churn'
if label_column in df.columns:
    label = df.pop(label_column)
    df[label_column] = label

# Encode target variable
df[label_column] = df[label_column].map({
    'Yes': 1, 'No': 0
}).astype(int)

# Identify feature types
numerical_cols = df.select_dtypes(
    include=['int64', 'float64']
).columns.tolist()

categorical_cols = df.select_dtypes(
    include=['object', 'category', 'bool']
).columns.tolist()

# Remove label from feature lists
numerical_cols = [
    col for col in numerical_cols
    if col != label_column
]

categorical_cols = [
    col for col in categorical_cols
    if col != label_column
]

# Apply Min-Max scaling to numerical features
scaler = MinMaxScaler()
df_numerical = pd.DataFrame(
    scaler.fit_transform(df[numerical_cols]),
    columns=numerical_cols
)

# One-hot encode contract feature
df_contract = pd.get_dummies(
    df[['Contract']],
    prefix='Contract'
)

# Encode remaining categorical variables
binary_cats = [
    col for col in categorical_cols
    if col != 'Contract'
```

```
]

df_binary = df[binary_cats].apply(
    lambda x: x.astype('category').cat.codes
)

# Combine all processed features
df_processed = pd.concat([
    df_numerical,
    df_binary,
    df_contract,
    df[label_column]
], axis=1)

# Save processed dataset
df_processed.to_csv(
    'Dataset_processed_for_ANN.csv',
    index=False
)
```

The implementation follows a structured preprocessing workflow. The target variable is first standardised into binary form. Feature types are then separated into numerical and categorical groups.

Numerical variables are scaled using Min-Max transformation to ensure consistent value ranges. Categorical variables are encoded using a hybrid approach. Contract features are one-hot encoded while other categorical variables are converted into integer codes.

Finally, all processed components are combined into a single dataset. This dataset is used for both ANN modelling and clustering analysis. The pipeline ensures consistency, reproducibility, and compatibility with machine learning algorithms.

Practical Impact

After scaling, all numerical features are standardised within the range of 0 to 1. This improves numerical stability during model training.

For neural networks, scaling helps gradient descent converge more efficiently. For clustering, scaling ensures that distance calculations are not biased toward features with larger values.

Therefore, scaling directly contributes to both model performance and interpretability.

5 Exploratory Data Analysis

Feature Distributions

The distribution plots reveal several important patterns. The `SeniorCitizen` variable is highly imbalanced, indicating that most customers are not senior citizens. The `tenure` variable spans a wide range, reflecting customers at different lifecycle stages. Monthly charges show a dispersed pattern, suggesting multiple pricing tiers. The `IsNewCustomer` feature is heavily skewed toward zero, indicating that only a small proportion of customers are newly acquired.

This imbalance is expected in real-world datasets. It highlights the importance of focusing on minority behavioural groups during modelling.

Correlation Analysis

The correlation heatmap shows that most features have weak pairwise relationships. This suggests low multicollinearity, which is beneficial for machine learning models. Strong negative correlations exist between contract variables due to one-hot encoding. This confirms that these variables represent mutually exclusive categories rather than independent predictors.

Low correlation across most features indicates that each variable contributes unique information. This improves model robustness and reduces redundancy.

Correlation with Churn

The correlation results highlight key behavioural drivers. Monthly charges show a positive relationship with churn, indicating that higher-paying customers may be more likely to leave. Tenure exhibits a negative relationship, suggesting that long-term customers are more stable. Contract-related features also contribute to churn patterns, reflecting the impact of commitment level.

Although correlations are not extremely high, they still provide useful directional insights. These patterns guide feature importance interpretation in later modelling stages.

Monthly Charges Distribution

The distribution of monthly charges displays multiple peaks. This suggests the presence of different service bundles or pricing levels. Such variation supports the idea that pricing structure plays a role in customer segmentation and churn behaviour.

The multimodal pattern also indicates that customers are grouped into distinct billing tiers. This aligns with the need for clustering analysis in later stages.

Charges vs Churn

The boxplot shows that churned customers generally have higher monthly charges. The median value for

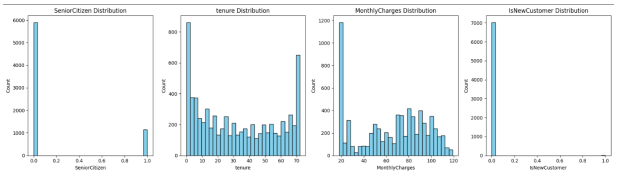

Figure 1: Distribution of key numerical and binary features

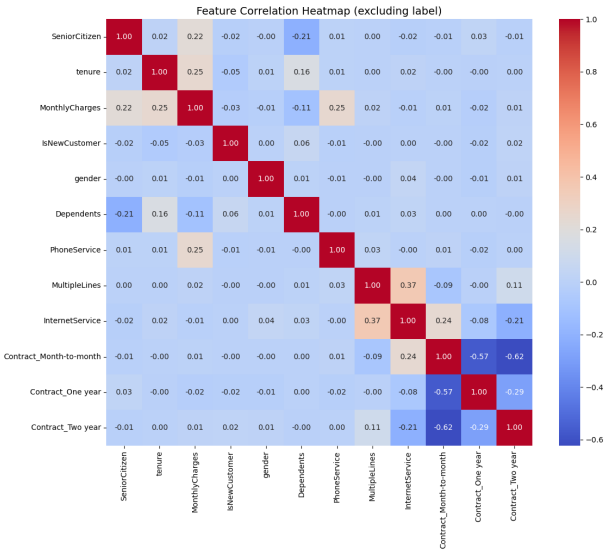

Figure 2: Feature correlation heatmap

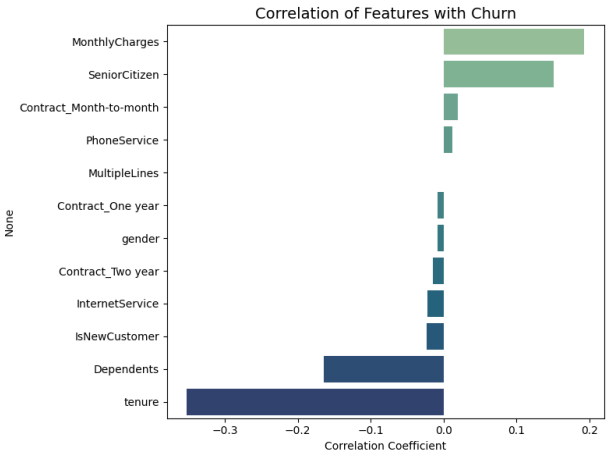

Figure 3: Correlation of features with churn

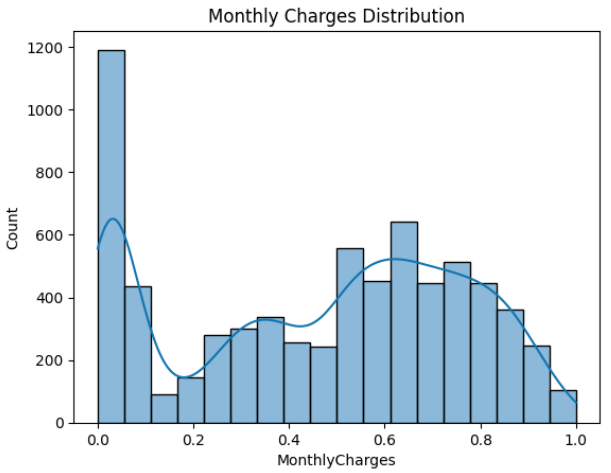

Figure 4: Monthly charges distribution

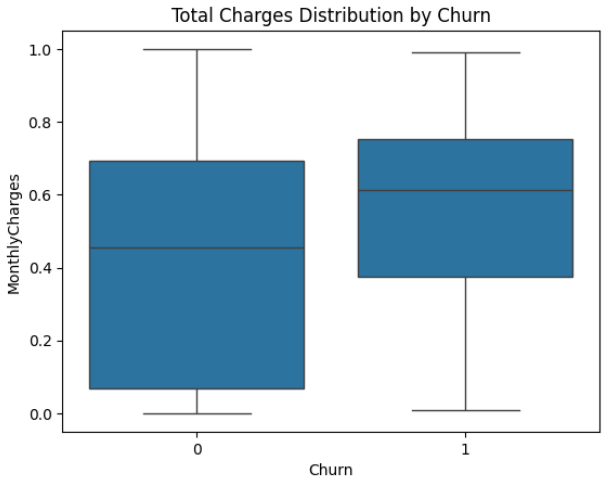

Figure 5: Monthly charges by churn status

churned customers is higher than that of retained customers. This indicates that pricing pressure may contribute to customer attrition. However, overlap between the two groups suggests that pricing is not the only factor influencing churn.

This result implies that churn is driven by multiple interacting factors. Pricing should therefore be considered alongside contract and service variables.

6 Data Distribution Summary

Table 1: Categorical Feature Distributions

Feature	**Distribution**
Gender	Male: 3555, Female: 3488
Dependents	No: 4933, Yes: 2110
PhoneService	Yes: 6361, No: 682
MultipleLines	No: 4072, Yes: 2971
InternetService	DSL: 3947, Fiber optic: 3096
Contract	Monthly: 3875, 1 year: 1473, 2 years: 1695
Churn	No: 5174, Yes: 1869

The categorical distribution confirms that most customers use phone services and monthly contracts. The churn distribution indicates moderate imbalance, which remains manageable for modelling.

7 Discussion

The preprocessing and exploratory analysis reveal that customer behaviour is influenced more by service and pricing factors than demographic variables. Contract structure and billing level appear to be key drivers of churn. The introduction of `IsNewCustomer` provides additional insight into early-stage customer behaviour.

The dataset shows low multicollinearity overall, which is beneficial for machine learning models. The observed distributions and correlations confirm that the dataset is suitable for both classification and clustering tasks.

The analysis also highlights that no single feature fully explains churn behaviour. This supports the use of machine learning models that can capture complex interactions between variables.

8 Conclusion

This report presented the preprocessing pipeline and exploratory analysis for customer churn prediction. The data was transformed through feature engineering, scaling, and encoding. Visual and statistical analysis confirmed the importance of contract type, tenure, and pricing in influencing churn.

The processed dataset provides a strong foundation for downstream modelling, including ANN classification and clustering-based segmentation. The structured preprocessing pipeline ensures that the dataset is both interpretable and model-ready.